Preface: Walking along the Shore

We are obsessed with ourselves. We study *our* history, *our* psychology, *our* philosophy, *our* gods. Much of our knowledge revolves around man himself, as if we were the most important thing in the universe. I think I like physics because it opens a window through which we can see further. It gives me the sense of fresh air entering the house.

What we see out there through the window is constantly surprising us. We have learned a great deal about the universe. In the course of the centuries we have come to realize just how very many wrong ideas we had. We thought that the Earth was flat, and that it was the still centre of our world. That the universe was small, and unchanging. We believed that man was a breed apart, without kinship to the other animals. We have learned of the existence of quarks, black holes, particles of light, waves of space, and of the extraordinary molecular structures in every cell of our bodies. The human race is like a growing child who discovers with amazement that the world consists not just of his bedroom and playground, but that it is vast, and that there are a thousand things to discover, and innumerable ideas quite different from those with which he began. The universe is multiform and boundless, and we continue to stumble upon new aspects of it. The more we learn about the world, the more we are amazed by its variety, beauty and simplicity.

But the more we discover, the more we understand that what we don't yet know is greater than what we know. The more powerful our telescopes, the stranger and more unexpected are the heavens we see. The closer we look at the minute detail of matter, the more we discover of its profound structure. Today we see almost to the Big Bang, the great explosion from which, 14 billion years ago, all the galaxies were born – but we have already begun to glimpse something beyond the Big Bang. We have learned that space is curved, but already foresee that this same space is woven from vibrating quantum grains.

Our knowledge of the elementary grammar of the world continues to grow. If we try to put together what we have learned about the physical world in the course of the twentieth century, the clues point towards something profoundly different from what we were taught at school. An elementary structure of the world is emerging, generated by a swarm of quantum events, where time and space do not exist. Quantum fields draw space, time, matter and light,